

- 1) If $u = 3 - 4i$ and $v = 4 - 3i$, find each of the following in the form $a + ib$ (4)
 - (a) $u + iv$
 - (b) vu
 - (c) $v^2 - u^2$
 - (d) $\frac{u}{v}$
- 2) Find the square roots of $-8 - 15i$ (3)
- 3) Show that $(x + 1 - i)$ is a factor of $x^3 - 2 - 2i$ (2)
- 4) If $z = x + iy$, find x and y when $\frac{2z}{1+i} - \frac{2z}{i} = \frac{5}{2+i}$ (3)
- 5)
 - (a) Given $z = x + iy$, express $\frac{z-i}{z+1}$ in the form $a + ib$ (3)
 - (b) If $\frac{z-i}{z+1}$ is real, find the equation of the locus of the point P representing z on an Argand diagram. (1)
- 6) If $z_1 = r_1 \text{cis} \theta_1$ and $z_2 = r_2 \text{cis} \theta_2$, prove that $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$ (2)
- 7) Find the modulus and principal argument of each of the complex numbers
 - (a) -3
 - (b) $-2i$
 - (c) $-\sqrt{3} + i$
- 8) Express $2 \text{cis} \left[-\frac{5\pi}{6} \right]$ in the form $x + iy$ (2)
- 9) Find the four fourth roots of $8(\sqrt{3} + i)$ (4)
- 10) Use De Moivre's Theorem to show that
 - (a) $(1 - i\sqrt{3})^9 = -512$ (3)
 - (b) $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ (3)

- 11) Draw neat diagrams to represent the following (6)

(a) $|z - 1 + i| = \sqrt{2}$

(b) $\left| \frac{z-1}{z+1} \right| \leq 1$

(c) $\text{Arg}(z+i) = -\frac{3\pi}{4}$

- 12) (a) Solve $z^3 = 1$, giving the complex roots in Mod-Arg form. (1)

- (b) If ω is one of the complex cube roots of unity, show that

(i) ω^2 is the other complex root. (1)

(ii) $\omega + \omega^2 = -1$ (1)

(iii) $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega) = 3$ (2)

- 13) On an Argand diagram P and Q represent the complex numbers z_1 and z_2 respectively. OPQ is an equilateral triangle in which $|z_1| = |z_2| = 1$.

- (a) Write an expression in terms of z_1 and z_2 for the complex number represented by the vector PQ . (1)

- (b) Find $|z_1 + z_2|$ (3)

- (c) Find a possible value of k if $\frac{z_1 + z_2}{z_1 - z_2} = k$ (2)

- 14) If $z = \cos \theta + i \sin \theta$, prove that

(a) $\frac{2}{z+1} = \frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta}$ and hence or otherwise prove that (2)

(b) $\frac{2}{z+1} = 1 - i \tan \frac{\theta}{2}$ (2)

13) If $15^\circ = 45^\circ - 30^\circ$,

(a) show that $\cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$,

(b) given $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$ and $z = \frac{1+\sqrt{3}i}{1+i}$, find

(i) $|z|$ (ii) $\text{Arg}(z)$

13) Given $z = \frac{1 - \cos 2\theta + i \sin 2\theta}{1 + \cos 2\theta - i \sin 2\theta}$, find

(a) $\text{Re}(z)$ (b) $\text{Arg}(z)$

(c) $|z|$ (d) $\bar{z}$

15) Find x in the domain $0 < x < \frac{\pi}{2}$, if $\frac{\sqrt{2}(\cos x - i \sin x)}{2+i} = \frac{1-3i}{5}$

EXT 2 ASSESS TASK #1 SOLUTIONS 2009

PART 1 Q1-5

1 a) $3-4i+i(4-3i)$
 $= 3-4i+4i+3$
 $= 6$ ①

b) $(3-4i)(4-3i)$
 $= 12-9i-16i-12$
 $= -25i$ ①

c) $(4-3i)^2 - (3-4i)^2$
 $= 16-24i-9-(9-24i-16)$
 $= 7-24i+7+24i$
 $= 14$ ①

d) $\frac{3-4i}{4-3i} \times \frac{4+3i}{4+3i}$
 $= \frac{12+9i-16i+12}{16+9}$
 $= \frac{24-7i}{25}$ ①

2, let $z^2 = -8-15i$
 $\therefore (x+iy)^2 = -8-15i$
 $x^2-y^2+2xyi = -8-15i$
 $\therefore x^2-y^2 = -8$ (1)
 $2xy = -15$ (2)

From (2) $y = \frac{-15}{2x}$

In (1) $x^2 - \frac{225}{4x^2} = -8$

$\therefore 4x^4 - 225 = -32x^2$

$4x^4 + 32x^2 - 225 = 0$

$(2x^2+25)(2x^2-9) = 0$

$\therefore x^2 = \frac{-25}{2} > \frac{9}{2}$

not real.

$\therefore x = \pm \frac{3}{\sqrt{2}}$ are only real solns.

$\therefore x = \pm \frac{3\sqrt{2}}{2}$ $y = \mp \frac{5\sqrt{2}}{2}$

$\therefore z = \frac{3\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i, -\frac{3\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$

OR use mod arg approach but be careful to write answers in exact form

i.e $z = \sqrt{17} \text{cis}(-90^\circ + \frac{1}{2} \tan^{-1}(\frac{15}{8}))$

or $\sqrt{17} \text{cis}(90^\circ + \frac{1}{2} \tan^{-1}(\frac{15}{8}))$

3, Easiest way is to use factor theorem (try to avoid long division!)

Let $P(x) = x^3 - 2 - 2i$

if $x+1-i$ is a factor

then $P(-1+i) = 0$

$P(-1+i) = (-1+i)^3 - 2 - 2i$

$= (-1+i)^2(-1+i) - 2 - 2i$

$= (1-2i-i)(-1+i) - 2 - 2i$

$= 2i+2-2-2i$

$= 0$ as req.

$\therefore x+1-i$ is a factor

OR could solve

$z^3 = 2+2i$

best way use mod-arg and verify that

$-1+i$ is a root

$\therefore x+1-i$ is a factor.

4, Many methods used but best way

$\frac{2z}{1+i} - \frac{2z}{i} = \frac{5}{2+i}$

Realise denom. individually

$\frac{2z}{1+i} \times \frac{1-i}{1-i} - \frac{2z}{i} \times \frac{i}{i} = \frac{5}{2+i} \times \frac{2-i}{2-i}$

$\frac{2z-2zi+2zi}{2} = \frac{10-5i}{5}$

$z - zi + 2zi = 2 - i$

$z + zi = 2 - i$

$z(1+i) = 2 - i$

$z = \frac{2-i}{1+i} \times \frac{1-i}{1-i}$

$= \frac{2-2i-i-1}{2}$

$= \frac{1-3i}{2}$

$= \frac{1}{2} - \frac{3}{2}i$

$\therefore x = \frac{1}{2}, y = -\frac{3}{2}$

5a) $\frac{z-i}{z+1} = \frac{z+iy-i}{x+iy+1}$

$= \frac{x+i(y-1)}{(x+1)+iy} \times \frac{(x+1)-iy}{(x+1)-iy}$

On simplifying:

$= \frac{x^2+x+y^2-y}{(x+1)^2+y^2} + \frac{i(y-x-1)}{(x+1)^2+y^2}$

b) If real (purely):

then $\text{Im}(z) = 0$

$\therefore y - x - 1 = 0$

$y = x + 1$ is

the locus - but excluding (1,0) and (0,1)

PART B Q6-10

6, $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

$\overline{z_1 z_2} = r_1 r_2 [\cos(\theta_1 + \theta_2) - i \sin(\theta_1 + \theta_2)]$

$\overline{z_1} \cdot \overline{z_2} = r_1 [\cos \theta_1 - i \sin \theta_1] \cdot r_2 [\cos \theta_2 - i \sin \theta_2]$

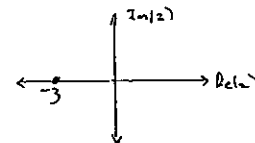
$= r_1 r_2 [\cos \theta_1 \cos \theta_2 - i \cos \theta_1 \sin \theta_2 - i \sin \theta_1 \cos \theta_2 + i^2 \sin \theta_1 \sin \theta_2]$

$= r_1 r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 - i (\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)]$

$= r_1 r_2 [\cos(\theta_1 + \theta_2) - i \sin(\theta_1 + \theta_2)]$

$= \overline{z_1 z_2}$

7, a)

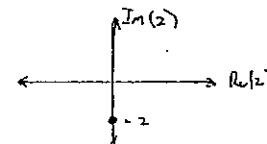


Let $z = -3 + 0i$

$|z| = 3$

$\text{Arg}(z) = \pi$ or 180°

b)

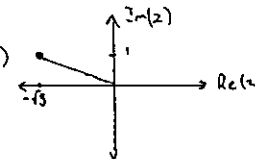


Let $z = 0 - 2i$

$|z| = 2$

$\text{Arg}(z) = -\frac{\pi}{2}$ or -90°

c)



Let $z = -\sqrt{3} + i$

$|z| = \sqrt{3^2 + 1^2} = 2$

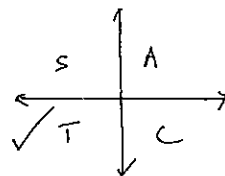
$\text{Arg}(z) = \frac{5\pi}{6}$ or 150°

8, $2 \text{cis}(-\frac{5\pi}{6}) = 2(\cos(-\frac{5\pi}{6}) + i \sin(-\frac{5\pi}{6}))$

or $2(\cos(-150^\circ) + i \sin(-150^\circ))$

$= 2(-\frac{\sqrt{3}}{2} + i(-\frac{1}{2}))$

$= -\sqrt{3} - i$

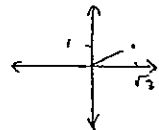


9, If $z^4 = 8(\sqrt{3} + i)$

$= 8 \cdot 2 \text{cis} \frac{\pi}{6}$

$= 2^4 \text{cis} \frac{\pi}{6}$ or $2^4 \text{cis} 30^\circ$

Let the roots be of the form $r \text{cis} \theta$



$$\therefore r^4 = 2^4$$

$$r = 2$$

$$4\theta = \frac{\pi}{6} + 2k\pi \quad k=0,1,2,3$$

$$\text{or } 4\theta = 360^\circ + 360^\circ \cdot k$$

$$\therefore \theta = \frac{\pi}{24} + \frac{2k\pi}{6} \quad \text{or } 7.5^\circ + 90^\circ \cdot k \quad k=0,1,2,3$$

$$\theta = \frac{\pi}{24}, \frac{13\pi}{24}, \frac{25\pi}{24}, \frac{37\pi}{24}$$

$$\text{or } 7.5^\circ, 97.5^\circ, 187.5^\circ, 277.5^\circ$$

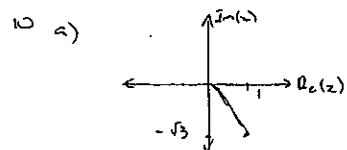
$$\text{Since } -\pi \leq \theta \leq \pi \quad \text{or } -180^\circ \leq \theta \leq 180^\circ$$

$$\theta = \frac{\pi}{24}, \frac{13\pi}{24}, \frac{25\pi}{24}, \frac{37\pi}{24}$$

$$\text{or } 7.5^\circ, 97.5^\circ, 187.5^\circ, 277.5^\circ$$

$$\therefore \text{Let } z = 2 \cos \frac{\pi}{24}, 2 \cos \frac{13\pi}{24}, 2 \cos \frac{25\pi}{24}, 2 \cos \frac{37\pi}{24}$$

$$\text{or } 2 \cos 7.5^\circ, 2 \cos 97.5^\circ, 2 \cos 187.5^\circ, 2 \cos 277.5^\circ$$



$$1 - i\sqrt{3} = 2 \cos -\frac{\pi}{3}$$

$$\therefore (1 - i\sqrt{3})^9 = \left(2 \cos -\frac{\pi}{3}\right)^9$$

$$= 2^9 \cos -3\pi$$

$$= 2^9 \cos -\pi$$

$$= 2^9 \cdot (-1 + 0i)$$

$$= 512 \cdot -1$$

$$= -512$$

b) $(\cos \theta)^3 = \cos 3\theta$

$$= \cos 3\theta + i \sin 3\theta \quad (1)$$

Also $(\cos \theta + i \sin \theta)^3 =$

$$= \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3\cos \theta i^2 \sin^2 \theta + i^3 \sin^3 \theta$$

$$= \cos^3 \theta - 3\cos \theta \sin^2 \theta + i (3\cos^2 \theta \sin \theta - \sin^3 \theta) \quad (2)$$

Equating the real parts of (1) & (2)

$$\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$$

$$= \cos^3 \theta - 3\cos \theta (1 - \cos^2 \theta)$$

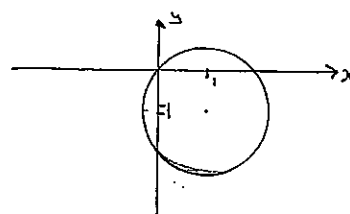
$$= \cos^3 \theta - 3\cos \theta + 3\cos^3 \theta$$

$$= 4\cos^3 \theta - 3\cos \theta$$

PART C Q11-14

Q11) a) $|z - 1 + i| = \sqrt{2}$

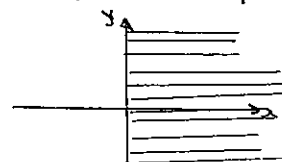
$$|z - (1 - i)| = \sqrt{2}$$



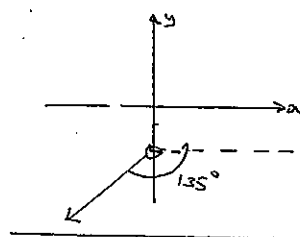
b) $\left| \frac{z-1}{z+1} \right| \leq 1$

$$\left| \frac{z-1}{z+1} \right| \leq 1$$

$$|z-1| \leq |z+1|$$



c) $\arg(z+i) = -135^\circ$



Q12) a) $z^3 = 1$

roots evenly spaced around the unit circle

$$z_1 = 1$$

$$z_2 = \text{cis } \frac{2\pi}{3}$$

$$z_3 = \text{cis } \frac{4\pi}{3}$$

b) Let $\omega = \text{cis } \frac{2\pi}{3}$

$$\therefore \omega^2 = (\text{cis } \frac{2\pi}{3})^2$$

$$= \text{cis } \frac{4\pi}{3}$$

but $\frac{4\pi}{3} = -\frac{2\pi}{3}$

$$\therefore \omega^2 = \text{cis } -\frac{2\pi}{3} \text{ which is the other root.}$$

ii) Let roots be $1, \omega, \omega^2$

Sum of the roots $= -\frac{b}{a}$

$$\therefore 1 + \omega + \omega^2 = 0$$

$$\omega + \omega^2 = -1$$

iii) $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega) = 3$

L.H.S.

$$= (1 + 2\omega + 3(-1-\omega))(1 + 2(-1-\omega) + 3\omega)$$

$$= (-2-\omega)(-1+\omega)$$

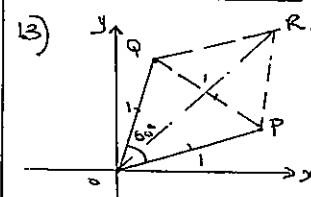
$$= 2 - 2\omega + \omega - \omega^2$$

$$= 2 - 2\omega + \omega - (-1-\omega)$$

$$= 2 - 2\omega + \omega + 1 + \omega$$

$$= 3$$

= R.H.S.



a) $PQ = z_2 - z_1$

b) $|z_1 + z_2| = \text{OR}$

$$OR^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos 120^\circ$$

$$= 3$$

$$OR = \sqrt{3}$$

c) $\frac{z_1 + z_2}{z_1 - z_2} = k$

$$\frac{z_1 + z_2}{z_1 - z_2} = k \left(\frac{z_1 - z_2}{z_1 - z_2} \right)$$

ie OR = k PQ

ie $\sqrt{3} = k$

but OPRQ is a rhombus

$\therefore$ the diagonals intersect at right angle

$\therefore PQ \perp OR$

a rotation through 90°

ie $k = i\sqrt{3}$

14) a) $\frac{z}{z+1} = \frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta}$

L.H.S. = $\frac{z}{z+1} \times \frac{(\cos \theta + i \sin \theta) - i \sin \theta}{(\cos \theta + i \sin \theta) - i \sin \theta}$

$$= \frac{2(\cos \theta + 1 - i \sin \theta)}{\cos^2 \theta + 2 \cos \theta + 1 + \sin^2 \theta}$$

$$= \frac{2(\cos \theta + 1 - i \sin \theta)}{2(1 + \cos \theta)}$$

$$= \frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta} = \text{R.H.S.}$$

b) $\frac{z}{z+1} = 1 - i \tan \frac{\theta}{2}$

L.H.S. = $\frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta}$

$$= \frac{1 + 2\cos^2 \frac{\theta}{2} - 1 - 2i \sin^2 \frac{\theta}{2}}{1 + 2\cos^2 \frac{\theta}{2} - 1}$$

$$= \frac{2(\cos^2 \frac{\theta}{2} - i \sin^2 \frac{\theta}{2})}{2\cos^2 \frac{\theta}{2}}$$

$$= \frac{\cos^2 \frac{\theta}{2}}{\cos^2 \frac{\theta}{2}} - \frac{i \sin^2 \frac{\theta}{2}}{\cos^2 \frac{\theta}{2}}$$

$$= 1 - i \frac{\sin^2 \frac{\theta}{2}}{\cos^2 \frac{\theta}{2}}$$

$$= 1 - i \tan \frac{\theta}{2}$$